

Introduction to DevOps and MLOps

Learning Objectives

- Understand why operationalizing machine learning (ML) is critical in the AI lifecycle and the key challenges in transitioning ML models from development to production.
- Summarize the concept of DevOps and MLOps, and their key features in operationalizing ML workflow.
- Explore how Azure Machine Learning (AzureML) simplifies MLOps, offering tools like pipelines for automation and MLflow for tracking and monitoring models.
- Apply AzureML pipelines to automate the end-to-end ML workflow, from data preparation and model training to model registration and deployment.

Agenda

- Business Context and Problem Statement
- Need of MLOps and DevOps
- Introduction of MLOps
- MLOps Lifecycle
- Introduction of DevOps
- MLOPs vs DevOps
- MLOps on Azure
- Conclusion

Business Context and Problem Statement

Leading healthcare
analytics company

XYZ Health Tech

Developed

Model

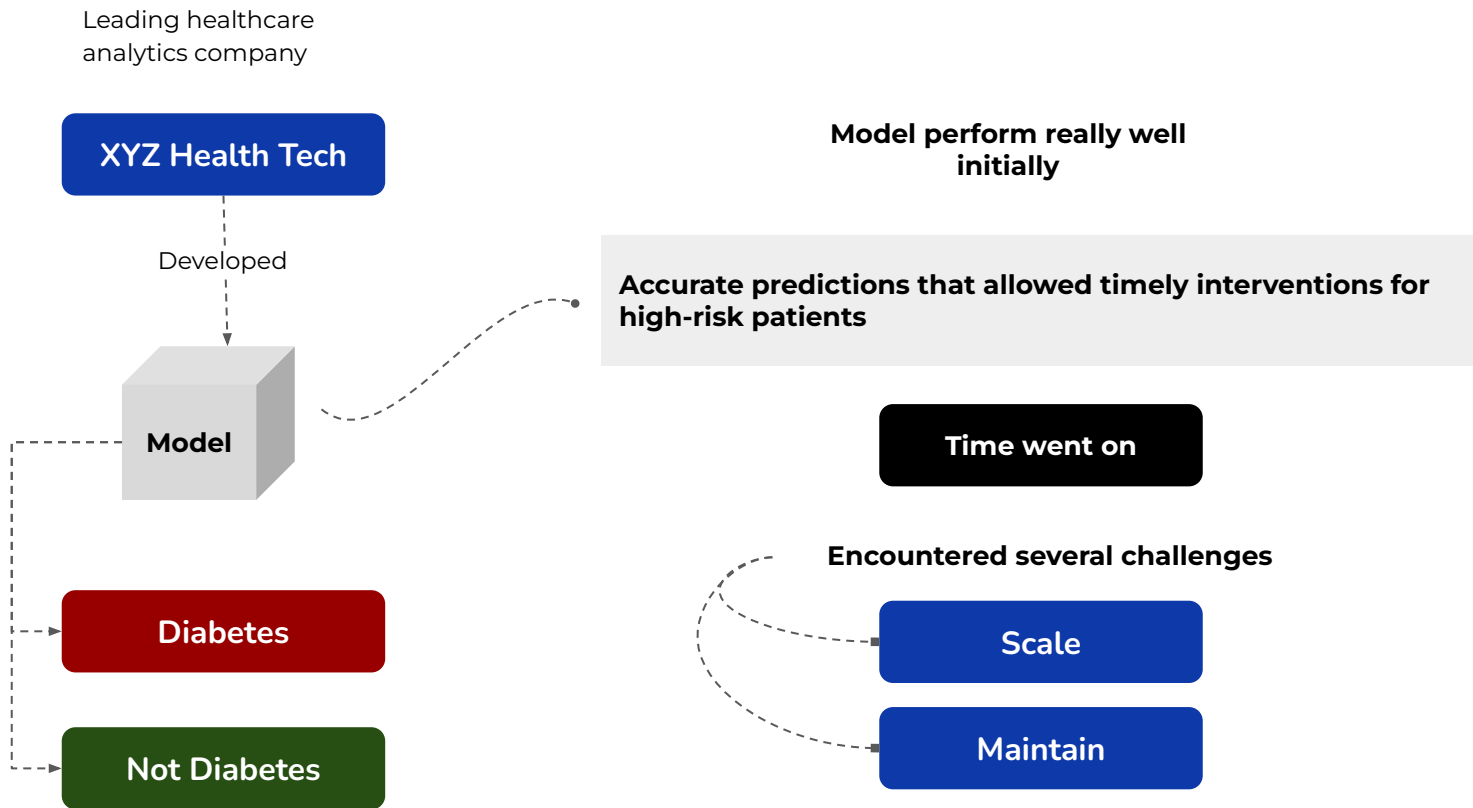
Model perform really well
initially

Accurate predictions that allowed timely interventions for
high-risk patients

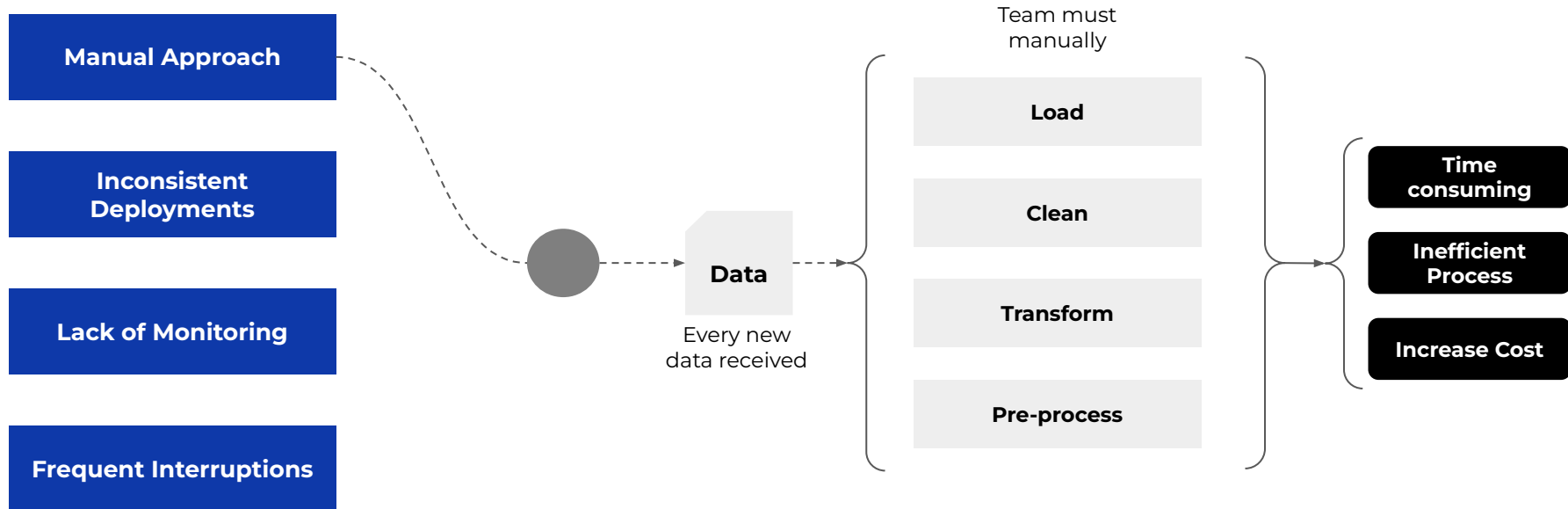
Diabetes

Not Diabetes

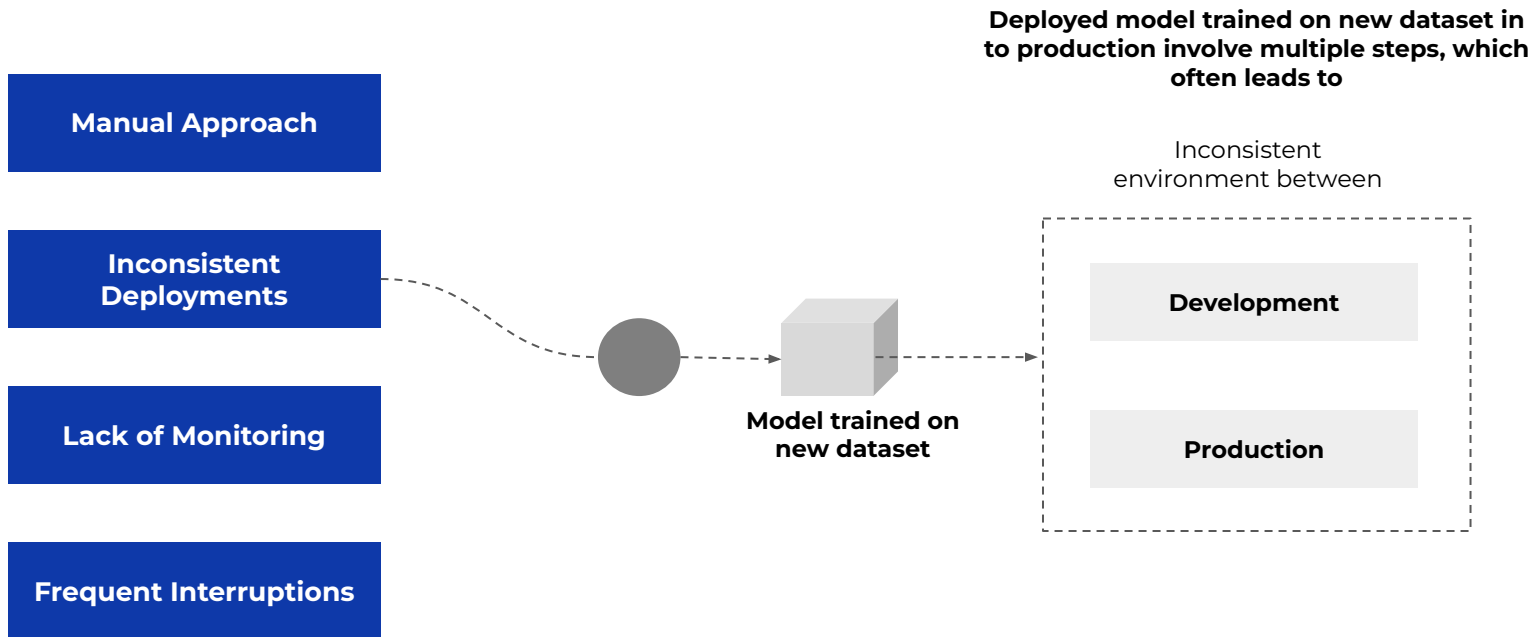
Business Context and Problem Statement



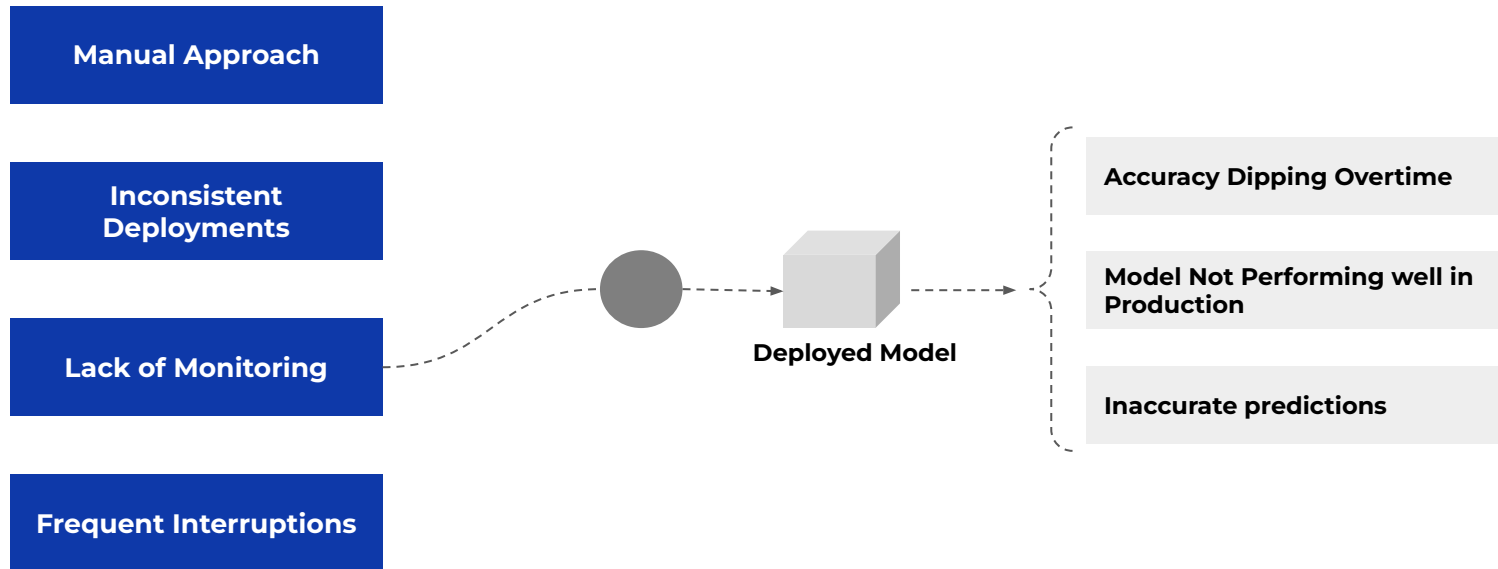
Challenges



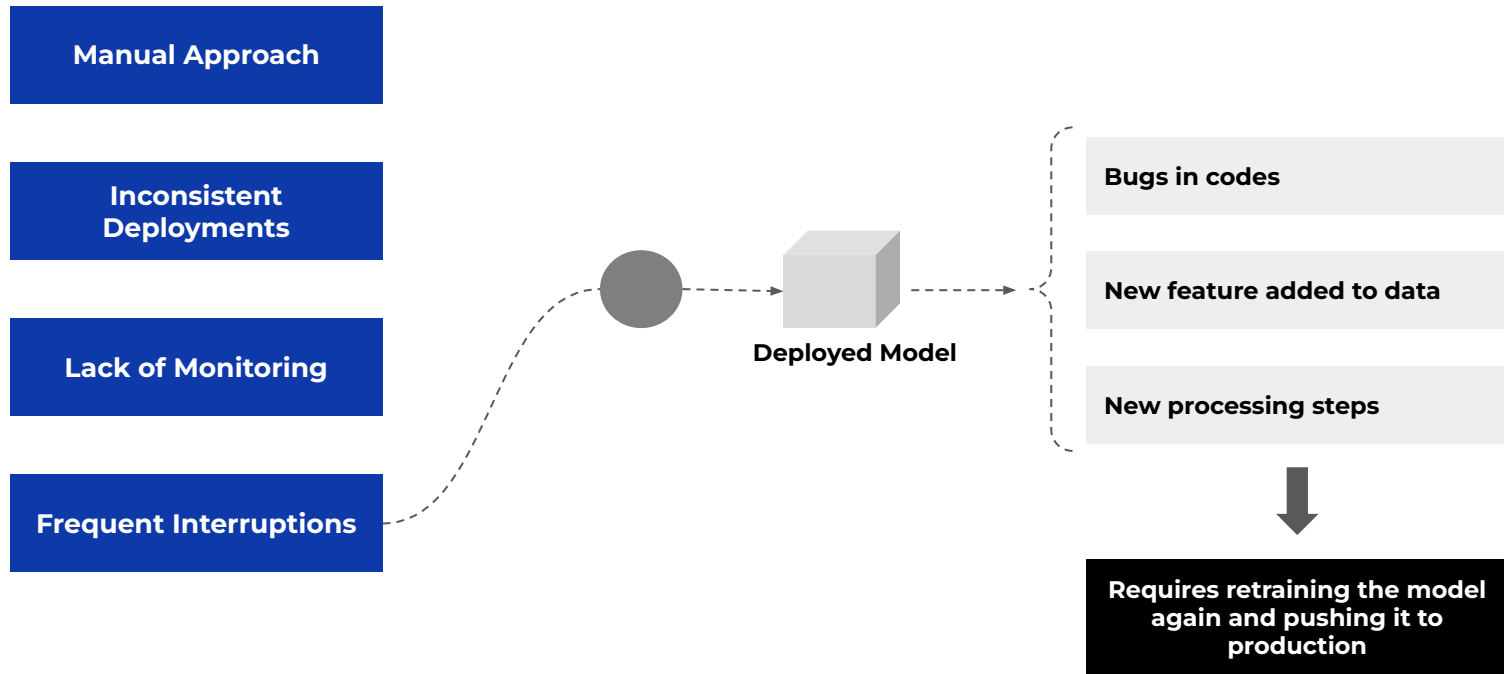
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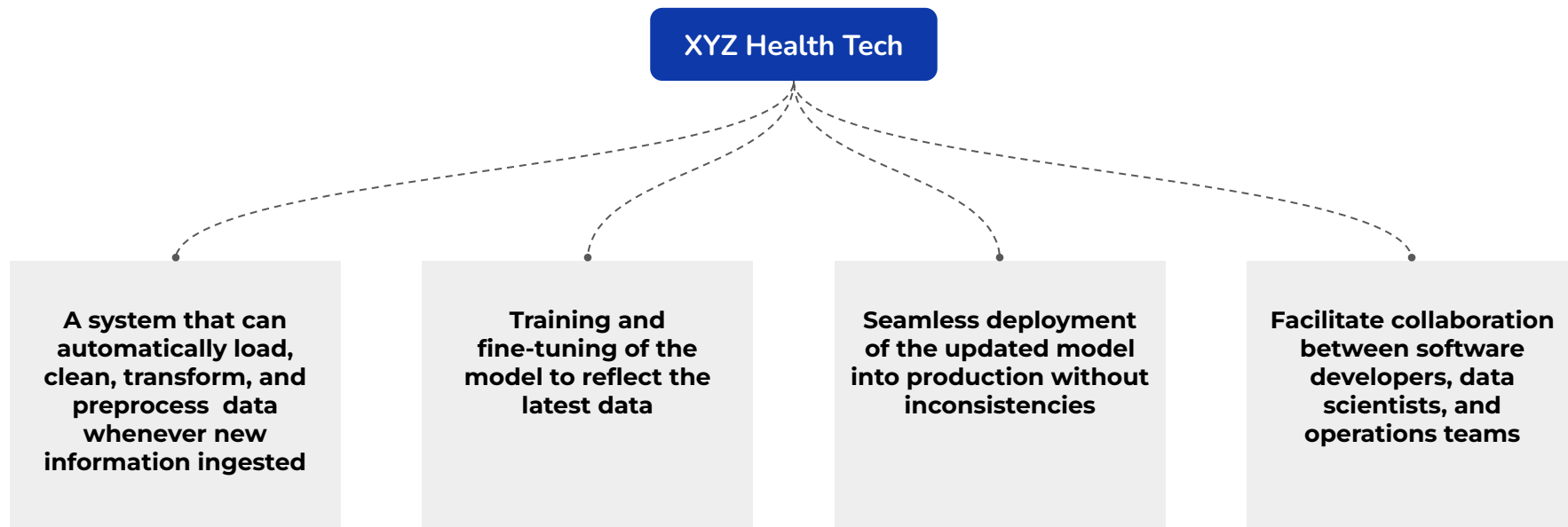
Challenges



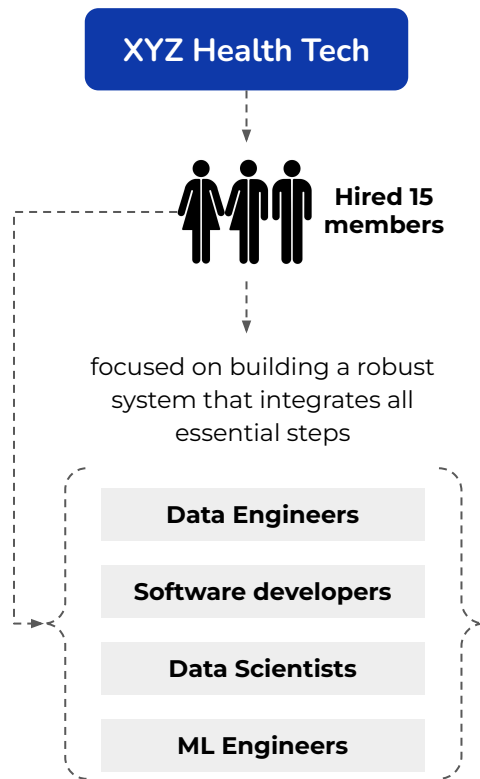
Challenges



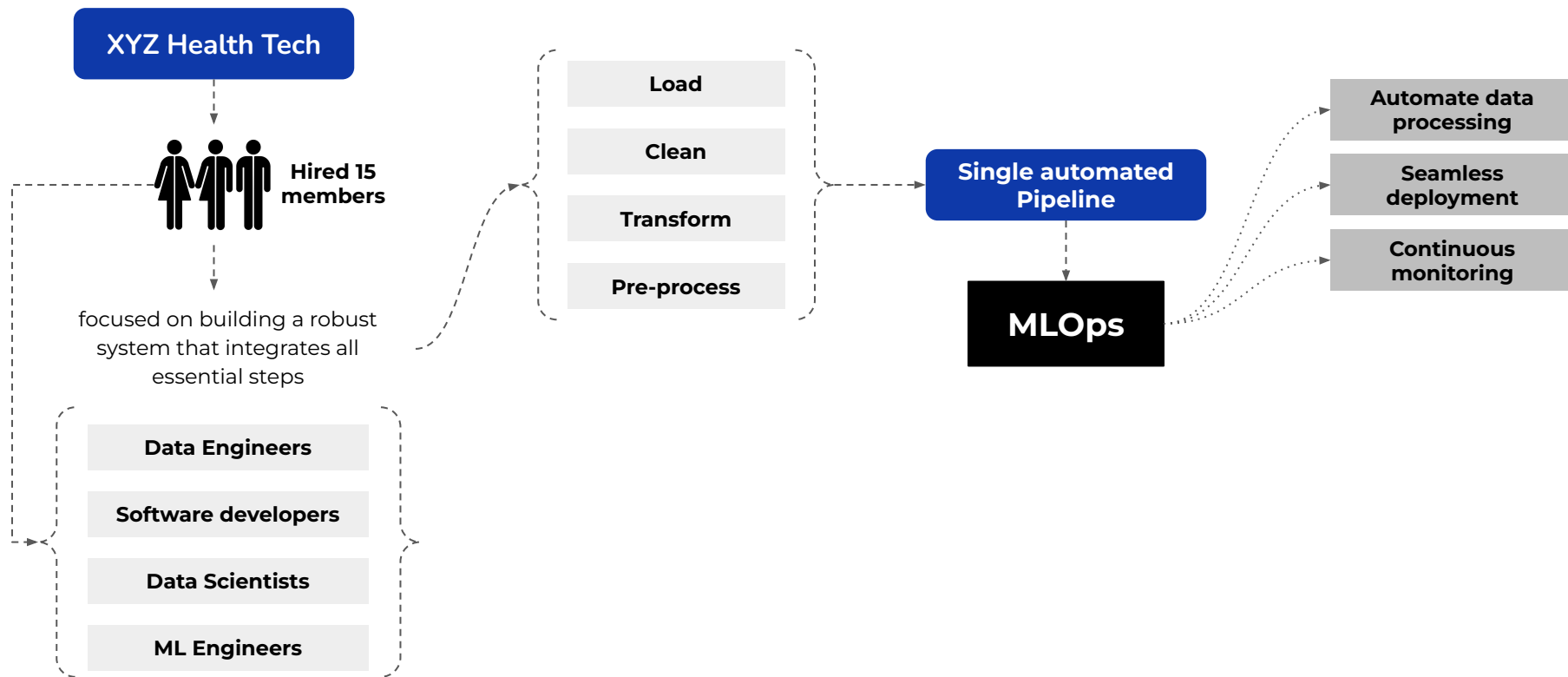
Needs of XYZ Health Tech



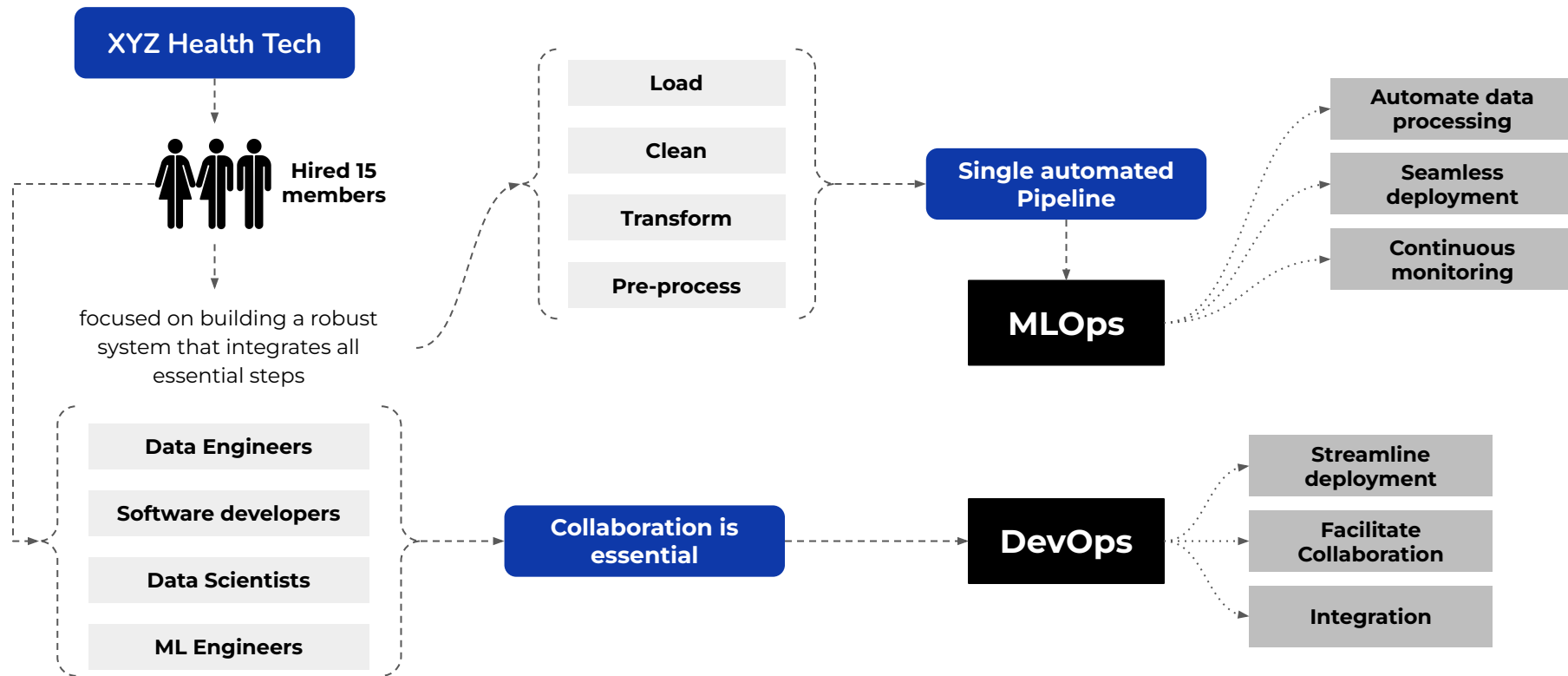
Need for MLOps and DevOps



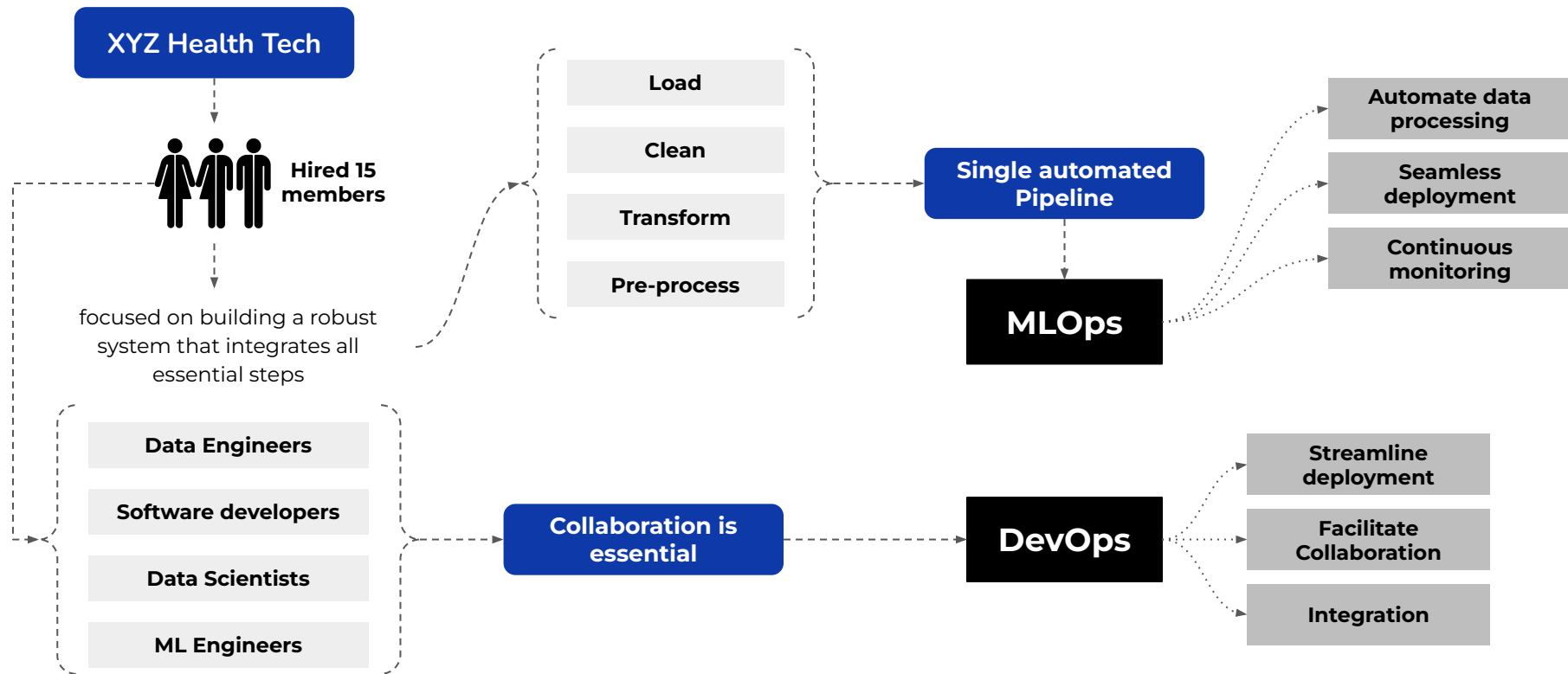
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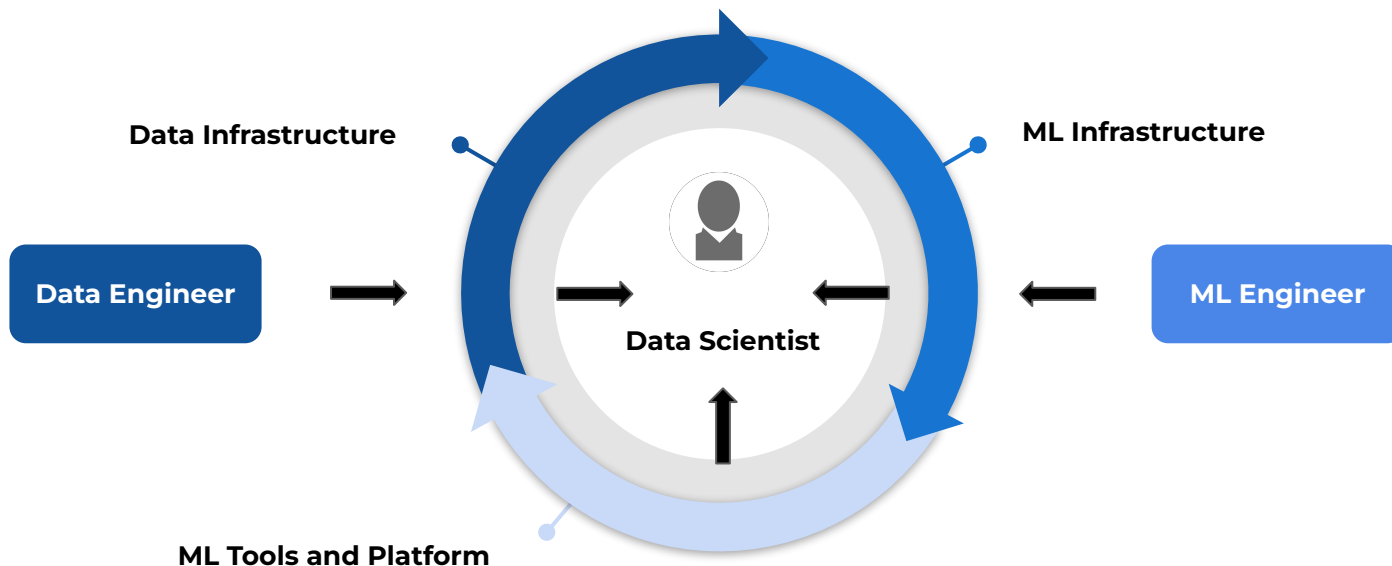
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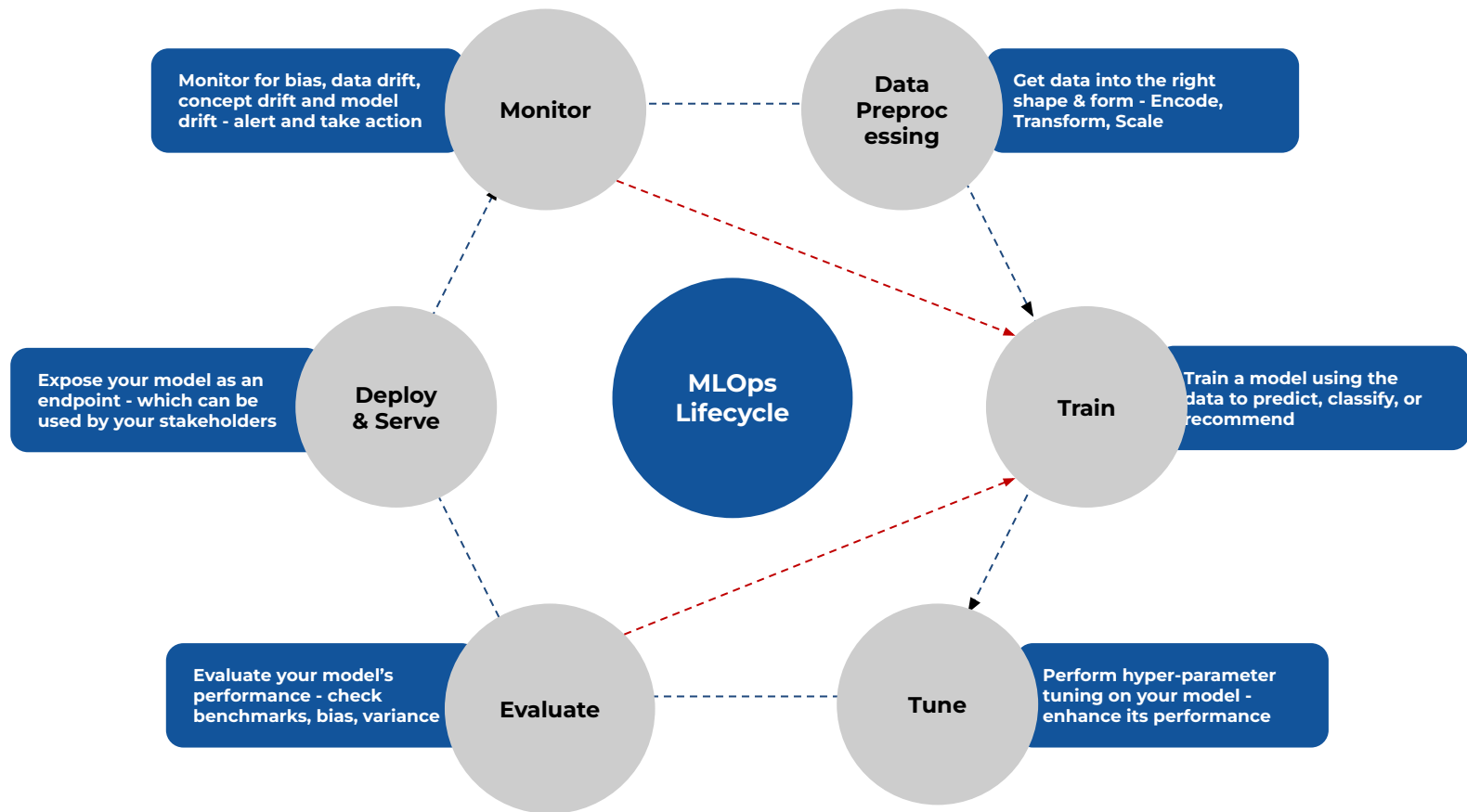
By implementing MLOps and DevOps, the team can automate repetitive tasks, foster efficient teamwork, and establish continuous integration and continuous deployment (CI/CD) pipelines

Introduction to MLOps

A cross-functional collaboration with one mission: **Operationalize Machine Learning**

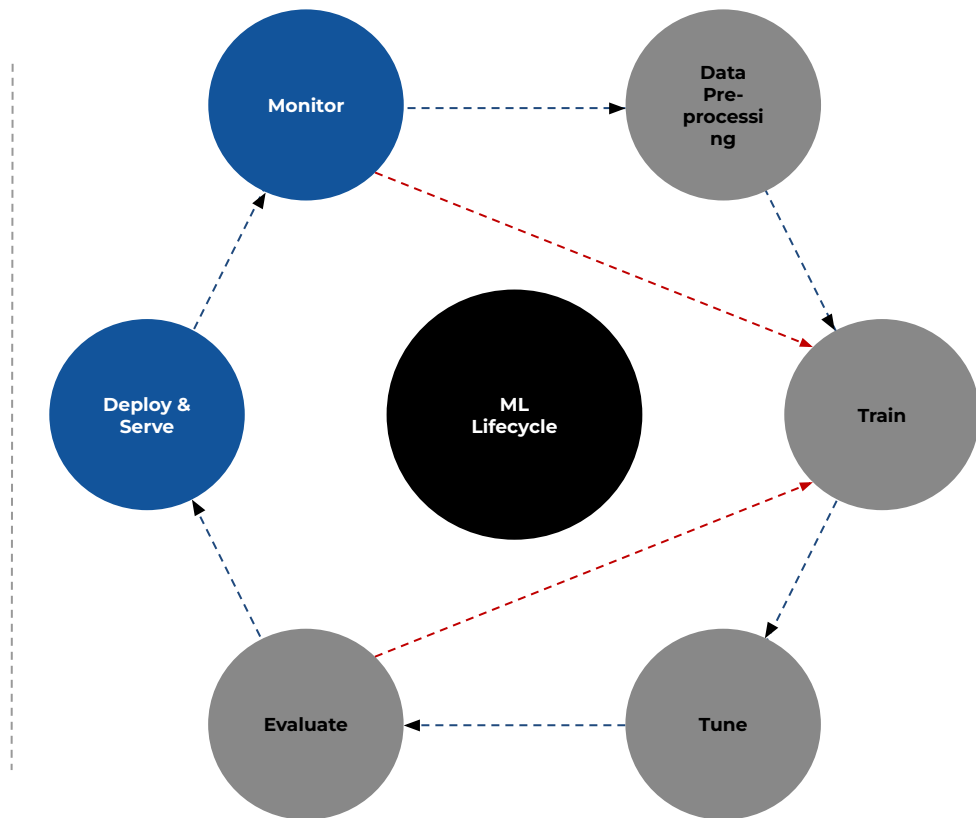


ML + Dev + Ops = MLOps

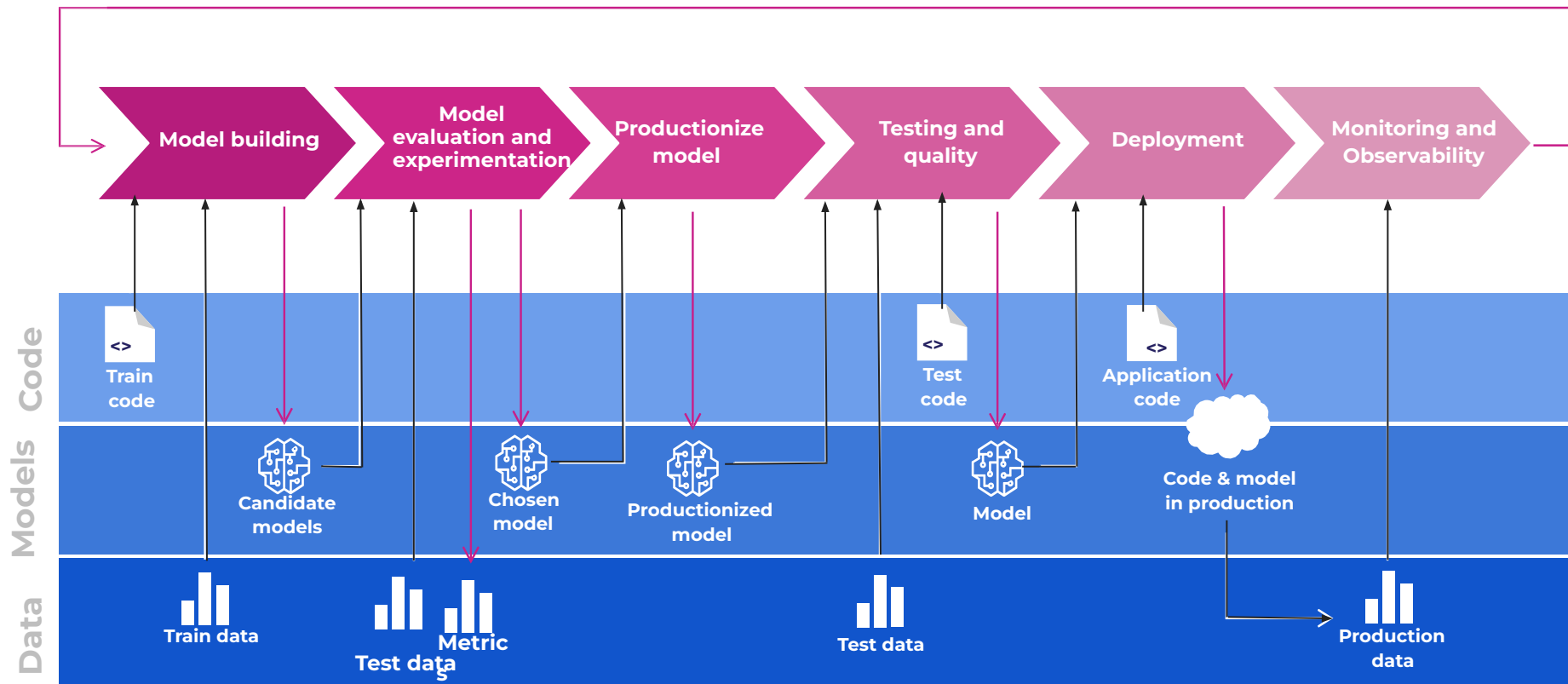


Need for MLOps in the Industry

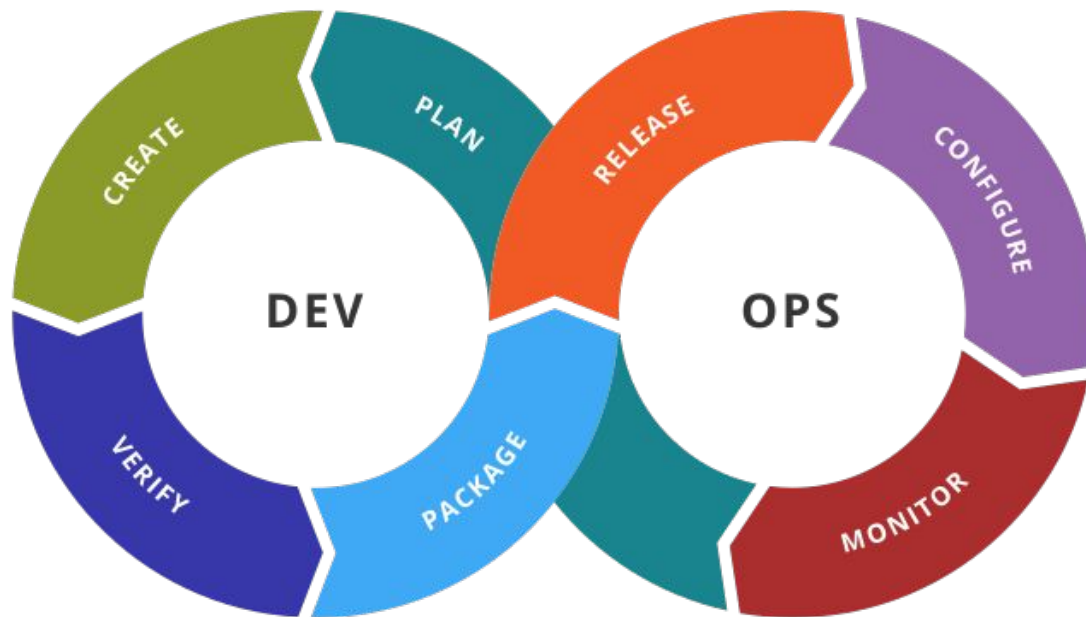
Data Science teams today need to go beyond just creating models



MLOps LifeCycle



Introduction to DevOps



Development + Operations = DevOps

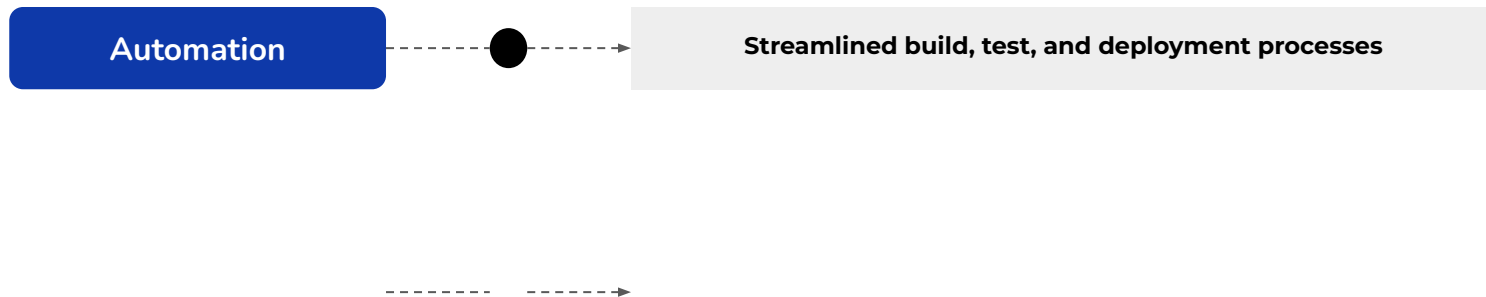
DevOps, a combination of "**Development**" and "**Operations**,"

- is a set of practices that automates and integrates software development and IT operations.
- with a goal to shorten the software development lifecycle while delivering high-quality software through:

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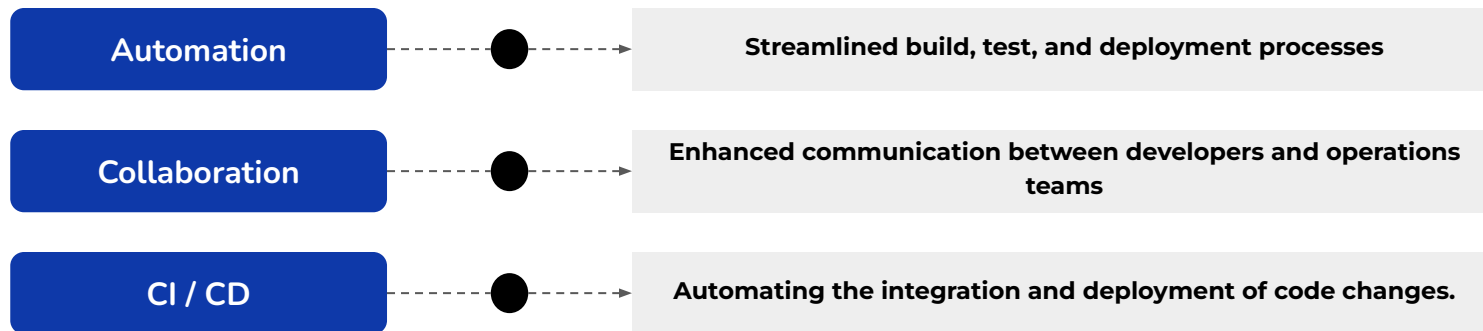
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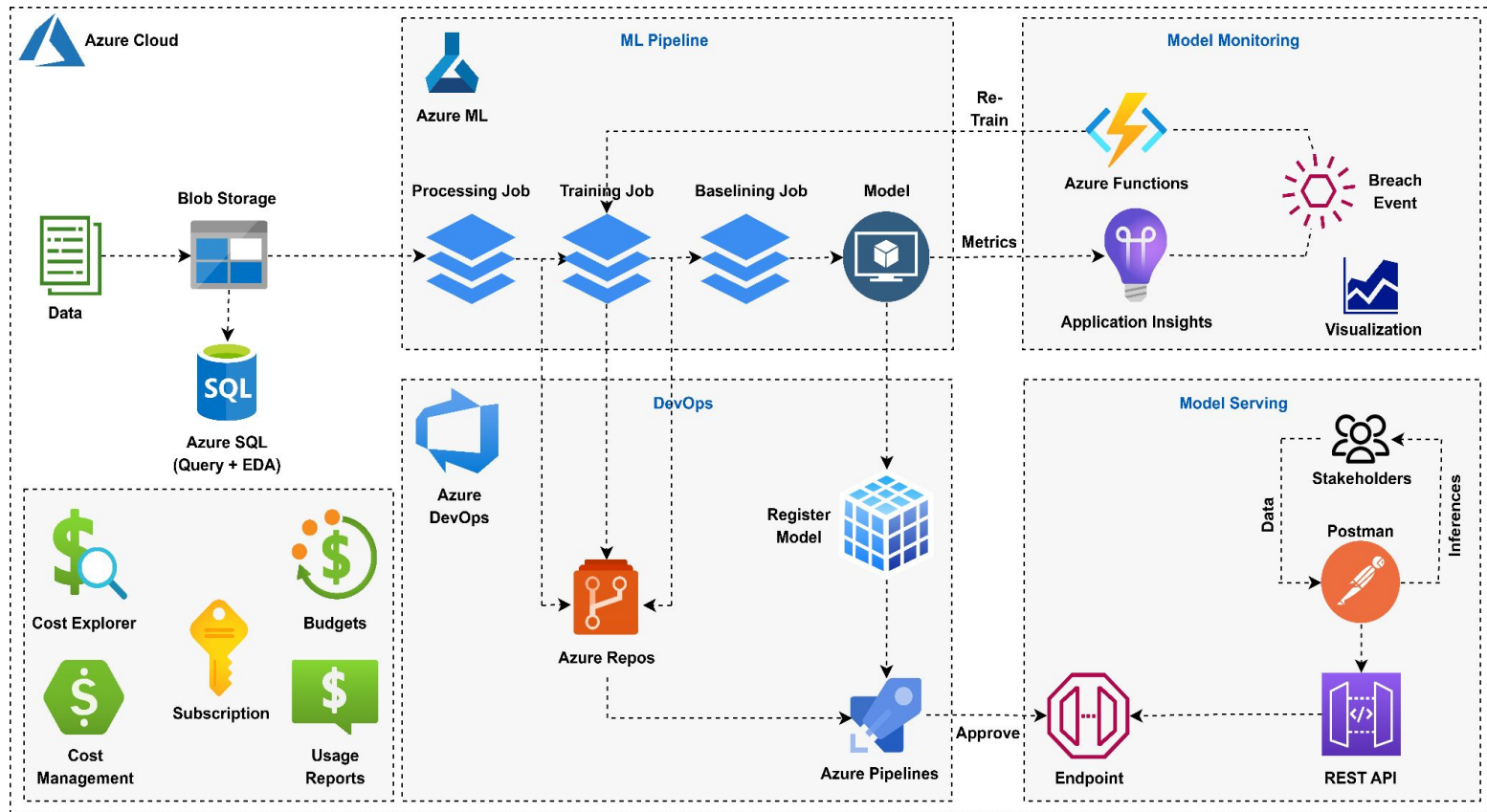
MLOps vs DevOps

	Devops	MLOps
Code versioning	✓	✓
Compute environment	✓	✓
Continuous integration/delivery (CI/CD)	✓	✓
Monitoring in production	✓	✓
Data provenance		✓
Datasets		✓
Models		✓
Metrics		✓
Workflows		✓

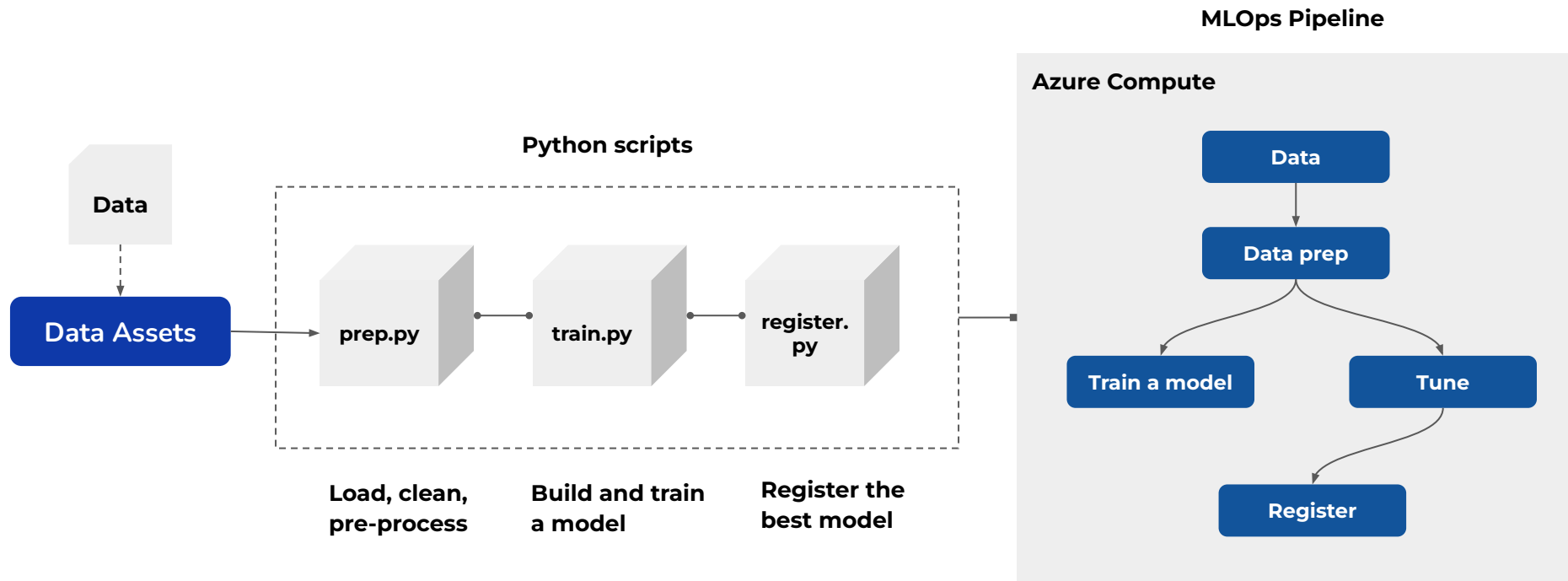
MLOPS

End-to-end ML
lifecycle
management

Azure ML for MLOps and DevOps



MLOps on Azure



Conclusion

- By adopting AzureML pipelines, MLflow, and MLOps principles, Nova HealthTech transforms its machine learning operations.
- The clinic now delivers reliable, scalable, and timely predictions, enabling better patient care and cementing its reputation as a leader in healthcare innovation.
- This success story demonstrates how AzureML and MLOps can revolutionize machine learning workflows, ensuring operational excellence and improved outcomes in healthcare and beyond.



Power Ahead!

